🌐 API List

🗺️ 1. Google Maps API

- Route generation
- Distance calculation
- Real-time navigation data
- Location services

🌦️ 2. Weather API

- Weather conditions (rain, storms, visibility)
- Used in risk calculation for route safety

📍 3. Places API

- Detects nearby:
  - Hospitals
  - Police stations
  - Public safe zones
- Used for emergency fallback routing

📡 4. Geolocation API

- Fetches real-time user location
- Tracks movement for dynamic rerouting

🔐 5. Firebase (optional but strong)

- User authentication
- Data storage (routes, preferences)
- Real-time updates (optional enhancement)

⚠️ Risk Scoring Explanation (Core AI Logic)

SURAKSHA evaluates each possible route using a **weighted safety scoring model** instead of only distance or time.

📊 Formula:

Risk=w1(Crime)+w2(Lighting)+w3(Isolation)+w4(Weather)+w5(Time)Risk = w_1(Crime) + w_2(Lighting) + w_3(Isolation) + w_4(Weather) + w_5(Time)Risk=w1(Crime)+w2(Lighting)+w3(Isolation)+w4(Weather)+w5(Time)

🧩 Factors Explained

1. Crime Risk

- Historical unsafe area data
- High-risk zones increase score

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2. Lighting Conditions

- Poorly lit streets = higher risk
- Well-lit urban roads = safer

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3. Isolation Level

- Empty/less crowded roads increase risk
- Busy public routes reduce risk

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4. Weather Conditions

- Rain, fog, storms increase difficulty and risk
- Clear weather reduces score

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5. Time Factor

- Night hours = higher risk
- Daytime = lower risk

🧮 Decision Logic

- Each route is scored independently
- The system compares all possible routes
- The **route with the lowest risk score is selected**, even if it is slightly longer